

MAR BASELIOS
INSTITUTE OF TECHNOLOGY AND SCIENCE
NELLIMATTOM P.O., KOTHAMANGALAM

PHONE : 0485 – 2859348, 2859318



LABORATORY RECORD

YEAR.....

Name:..... Roll No:.....

Branch: Batch:



MAR BASELIOS
INSTITUTE OF TECHNOLOGY AND SCIENCE
NELLIMATTOM P.O., KOTHAMANGALAM
PHONE : 0485 – 2859348, 2859318

LABORATORY RECORD

YEAR.....

Name:..... Roll No:.....

Branch: Batch:

*Certified that this is the Bonafide Record of Practical work done in the
..... Laboratory Mar Baselios Institute of Technology and
Science Nellimattom P.O, Kothamangalam, by*

Head of the Department

Staff Member in Charge

Uni. Ex. Reg:No.....of March/Sept.....

Uni. Ex. Reg:No.....of March/Sept.....

Internal Examiner

External Examiner

INDEX

S.No.	Date	PROGRAM	Page No.	Signature of Staff In-Charge
CO 1 : UNDERSTAND OBJECT-ORIENTED CONCEPTS AND DESIGN CLASSES AND OBJECTS TO SOLVE PROBLEMS				
1.		Define a class 'product' with data members pcode, pname and price. Create 3 objects of the class and find the product with the lowest price.	1	
2.		Read two matrices from the console and perform matrix addition.	4	
3.		Add complex numbers.	7	
4.		Read a matrix from the console and check whether it is symmetric or not.	9	
5.		Create CPU with attribute price. Create an inner class Processor (no. of cores, manufacturer) and static nested class RAM (memory, manufacturer). Create an object of CPU and print information of Processor and RAM.	11	
CO 2 : IMPLEMENT ARRAYS AND STRINGS				
6.		Program to Sort Strings.	14	
7.		Search an Element in an Array.	17	
8.		Perform String Manipulations.	19	
9.		Program to Create a Class for Employee and Search for an Employee Given eNo, using the concept of Array of Objects.	21	
CO 3 : IMPLEMENT OBJECT-ORIENTED CONCEPTS LIKE INHERITANCE, OVERLOADING AND INTERFACES				
10.		Area of different shapes using overloaded functions.	24	
11.		Create a class 'Employee' and another class 'Teacher' that inherit the properties of Employee class. Use array of objects to display details of N teachers.	27	

12.		Create a class 'Person' and another class 'Employee' that inherits the properties of class Person. Use array of objects to display details of N teachers.	31	
13.		Write a program that has classes Publisher, Book, Literature and Fiction. Read the information and print the details of books from either the category, using inheritance.	37	
14.		Create classes Student and Sports. Create another class Result inherited from Student and Sports. Display academic and sports score of a student.	42	
15.		Create an Interface with prototypes of functions area() and perimeter(). Create a menu driven program to find area and perimeter of objects.	45	
16.		Prepare Bill with the Given Format using calculate method from Interface.	48	
CO 4 : IMPLEMENT PACKAGES, EXCEPTION HANDLING, MULTITHREADING, AND GENERIC PROGRAMMING USING JAVA.UTIL PACKAGE AND COLLECTION FRAMEWORK				
17.		Create a Graphics package that has classes and interfaces for figures Rectangle, Triangle, Square and Circle. Test the package by finding the area of these figures and implementing all operations.	51	
18.		Create an Arithmetic package that has classes and interfaces for the 4 basic arithmetic operations. Test the package by implementing all operations on two given numbers.	54	
19.		Write a user defined exception class to authenticate the user name and password.	57	
20.		Find the average of N positive integers, raising a user defined exception for each negative input.	59	
21.		Define 2 classes; one for generating multiplication table of 5 and other for displaying first N prime numbers. Implement using threads. (Thread class)	62	
22.		Define 2 classes; one for generating Fibonacci numbers and other for displaying even numbers in a given range. Implement using threads. (Runnable Interface)	65	
23.		Producer/Consumer using IPC.	68	
24.		Program to create a generic stack and do the Push and Pop operations.	71	

25.		Using generic method perform Bubble sort.	74	
CO 5 : DEVELOP APPLICATIONS TO HANDLE EVENTS USING APPLETS				
26.		Program to Draw Circle, Rectangle, and Line in Applet.	76	
27.		Program to Find Maximum of three numbers using AWT.	78	
28.		Find the Percentage of marks obtained by a student in 5 subjects. Display a happy face if he secures above 50% or a sad face if otherwise.	81	
29.		Implement a simple calculator using AWT components.	84	
CO 6 : DEVELOP APPLICATIONS USING FILES AND NETWORKING CONCEPTS				
30.		Program to List the sub directories and files in a given directory and also search for a file name.	87	
31.		Write a program to write to a file, then read from the file and display the contents on the console.	90	
32.		Write a program to copy one file to another.	92	
33.		Write a program that reads from a file having integers. Copy even numbers and odd numbers to separate files.	94	
34.		Client-Server Communication using Socket – TCP/IP.	97	
35.		Client-Server Communication using Datagram Socket (UDP)	100	

Experiment – 1**Aim :**

Define a class 'product' with data members pcode, pname and price. Create 3 objects of the class and find the product having the lowest price.

Program Code :

```
public class product
{
    int pcode;
    String pname;
    double price;
    double lowest;
    void data(int c,String n,double p){
        pcode = c;
        pname = n;
        price = p;
    }
    void display(){
        System.out.println(pcode+"\t\t"+pname+"\t\t"+price);
    }
    static void findLowest(double price1,double price2,double price3){
        if(price1<=price2 && price1<=price3){
            System.out.println("\n Product 1 is of the lowest price!");
        }
        else if(price2<=price1 && price2<=price3){
            System.out.println("\n Product 2 is of the lowest price!");
        }
        else{
            System.out.println("\n Product 3 is of the lowest price!");
        }
    }
    public static void main(String args[]){
        product obj1 = new product();
```

```
        product obj2 = new product();
        product obj3 = new product();
        obj1.data(101,"Product_1",100.0);
        obj2.data(102,"Product_2",128.40);
        obj3.data(103,"Product_3",790.00);
        System.out.println("Product Information:\nProduct_code\tProduct_name\tProduct_price");
        obj1.display();
        obj2.display();
        obj3.display();
        findLowest(obj1.price,obj2.price,obj3.price);
    }
}
```

Output :

Product Information:

Product_code	Product_name	Product_price
101	Product_1	100.0
102	Product_2	128.4
103	Product_3	790.0

Product 1 is of the lowest price!

Experiment – 2

Aim :

Read 2 matrices from the console and perform matrix addition.

Program Code :

```
import java.util.Scanner;

public class add_matrix {

    public static void main(String args[]) {

        int row, col, i, j;

        Scanner in = new Scanner(System.in);

        System.out.println("Enter the number of rows: ");

        row = in.nextInt();

        System.out.println("Enter the number of columns: ");

        col = in.nextInt();

        int mat1[][] = new int[row][col];

        int mat2[][] = new int[row][col];

        int res[][] = new int[row][col];

        System.out.println("Enter the elements of matrix 1");

        for (i = 0; i < row; i++) {

            for (j = 0; j < col; j++) {

                mat1[i][j] = in.nextInt();

            }

            System.out.println();

        }

        System.out.println("Enter the elements of matrix 2");

        for (i = 0; i < row; i++) {

            for (j = 0; j < col; j++) {

                mat2[i][j] = in.nextInt();

            }

            System.out.println();

        }

        for (i = 0; i < row; i++) {

            for (j = 0; j < col; j++) {
```

```
        res[i][j] = mat1[i][j] + mat2[i][j];
    }
}
System.out.println("Sum of matrices:");
for (i = 0; i < row; i++) {
    for (j = 0; j < col; j++) {
        System.out.print(res[i][j] + "\t");
    }
    System.out.println();
}
}
```

Output :

Enter the number of rows:

2

Enter the number of columns:

3

Enter the elements of matrix 1

1 2 3

4 5 6

Enter the elements of matrix 2

7 8 9

1 2 3

Sum of matrices:

8 10 12

5 7 9

Experiment – 3

Aim :

Add complex numbers.

Program Code :

```
public class complex {  
    int r;  
    int i;  
    complex(int real,int img) {  
        r = real;  
        i = img;  
    }  
    void display() {  
        System.out.println(r+" "+i+"i");  
    }  
    static void add(int r1,int i1,int r2,int i2) {  
        r1 = r1 + r2;  
        i1 = i1 + i2;  
        System.out.println("After Addition = "+r1+" "+i1+"i");  
    }  
    public static void main(String args[]) {  
        complex first = new complex(5,4);  
        complex second = new complex(7,9);  
        System.out.println("Complex numbers are:");  
        first.display();  
        second.display();  
        add(first.r,first.i,second.r,second.i);  
    }  
}
```


Output :

Complex numbers are:

$$5+4i$$

$$7+9i$$

$$\text{After Addition} = 12+13i$$

Experiment – 4**Aim :**

Read a matrix from the console and check whether it is symmetric or not.

Program Code :

```
import java.util.Scanner;

public class Symmetric {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the Number of rows of the Matrix:");

        int row = sc.nextInt();

        System.out.println("Enter the Number of Columns of the Matrix:");

        int col = sc.nextInt();

        int[][] matrix = new int[row][col];

        int i,j;

        boolean state = true;

        for(i = 0; i < row; i++) {

            for(j = 0; j < col; j++) {

                System.out.println("Enter the Element at M(" + i + ", " + j + ")");

                matrix[i][j] = sc.nextInt();

            }

        }

        for(i = 0; i < row; i++) {

            for(j = 0; j < col; j++) {

                if(matrix[i][j] != matrix[j][i]) {

                    state = false;

                    break;

                }

            }

        }

        if(state) {

            System.out.println("Matrix is Symmetric");

        } else {

            System.out.println("Matrix is NOT Symmetric");

        }

    }

}
```

Output :

Enter the Number of rows of the Matrix:

3

Enter the Number of Columns of the Matrix:

3

Enter the Element at M(0,0):

1

Enter the Element at M(0,1):

2

Enter the Element at M(0,2):

6

Enter the Element at M(1,0):

2

Enter the Element at M(1,1):

8

Enter the Element at M(1,2):

4

Enter the Element at M(2,0):

6

Enter the Element at M(2,1):

4

Enter the Element at M(2,2):

5

Matrix is Symmetric

Experiment - 5**Aim :**

Create CPU with attribute price. Create inner class Processor (no. of cores, manufacturer) and static nested class RAM (memory, manufacturer). Create an object of CPU and print information of Processor and RAM.

Program Code :

```
public class cpu{
    int price;
    class processor{
        int cores;
        String producer;
        processor(int noC, String manu){
            cores=noC;
            producer=manu;
        }
        void display(){
            System.out.println("\nProcessor info");
            System.out.println("No. of Cores = "+cores);
            System.out.println("Manufacturer = "+producer+"\n");
        }
    }
    static class ram{
        int mem;
        String manuf;
        ram(int memory,String producer ){
            mem=memory;
            manuf=producer;
        }
        void display(){
            System.out.println("\nRAM info");
            System.out.println("Memory = "+mem+" GB");
            System.out.println("Manufacturer = "+manuf+"\n");
        }
    }
}
```

```
    }}  
    public static void main(String[] args) {  
        cpu.ram obj1= new cpu.ram(8,"Intel");  
        cpu obj2 = new cpu();  
        cpu.processor obj3 = obj2.new processor(8,"Samsung");  
        obj1.display();  
        obj3.display();  
    }  
}
```

Output :

RAM info

Memory = 8 GB

Manufacturer = Intel

Processor info

No. of Cores = 8

Manufacturer = Samsung

Experiment - 6

Aim :

Program to Sort strings.

Program Code :

```
import java.util.Scanner;

public class sort_string {

    public static void main(String args[]) {

        String temp;

        int i,j,count;

        Scanner scan = new Scanner(System.in);

        System.out.println("Enter number of strings you would like to enter:");

        count = scan.nextInt();

        String[] str = new String[count];

        Scanner scan2 = new Scanner(System.in);

        System.out.println();

        System.out.println("Enter the strings one by one:");

        for(i=0;i<count;i++)

            {

                str[i] = scan2.nextLine();

            }

        System.out.println();

        System.out.println("The strings before sorting:");

        for(i=0;i<count;i++)

        {

            System.out.println(str[i]);

        }

        for(i=0;i<count;i++)

        {

            for(j=i+1;j<count;j++)

            {

                if((str[i].compareTo(str[j]))>0)

                {
```

```
        temp = str[i];
        str[i] = str[j];
        str[j] = temp;
    }
}
}

System.out.println();
System.out.println("The strings after sorting:");
for(i=0;i<count;i++)
{
    System.out.println(str[i]);
}
}
}
```


Output :

Enter number of strings you would like to enter:

3

Enter the strings one by one:

yellow

blue

red

The strings before sorting:

yellow

blue

red

The strings after sorting:

blue

red

yellow

Experiment - 7**Aim :**

Search an element in an array.

Program Code :

```
import java.util.*;

public class elementSearch {

    public static void main(String args[]) {

        int count,num,i;

        int[] inputArray = new int[500];

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the number of elements to be inserted:");

        count = sc.nextInt();

        System.out.println("Start to enter those "+ count +" elements:");

        for(i=0;i<count;i++) {

            inputArray[i] = sc.nextInt();

        }

        System.out.println("Enter the element to be searched:");

        num = sc.nextInt();

        for(i=0;i<count;i++) {

            if(num == inputArray[i]) {

                System.out.println(num + " is present at index "+i+"");

                break;

            }

        }

        if(i==count)

            System.out.println(num + " is not present in the array");

    }

}
```

Output :

Enter the number of elements to be inserted:

3

Start to enter those 3 elements:

23

45

66

Enter the element to be searched:

45

45 is present at index 1

Experiment - 8**Aim :**

Perform string manipulations.

Program Code :

```
public class Strings
{
    public static void main(String[] args)
    {
        String text="Object Oriented Programming";
        System.out.println("Original Text: "+text);
        System.out.println("\nThe given text is empty -> "+text.isEmpty());
        System.out.println("\nLength of the given text: "+text.length());
        System.out.println("\nTo Uppercase: "+text.toUpperCase());
        System.out.println("\nTo Lowercase: "+text.toLowerCase());
        System.out.println("\nSubstring: "+text.substring(7));
        System.out.println("\nSubsequence: "+text.subSequence(7,15));
        System.out.println("\nReplace O with $: "+text.replace('O','$'));
        System.out.println("\nReplace all text with 'Java Lab': "+text.replaceAll(text,"Java Lab"));
        System.out.println("\nReplace first word: "+text.replaceFirst("Object","Java"));
        System.out.println("\nCharacter at index 8: "+text.charAt(8));
        int ascii=text.charAt(8);
        System.out.println("ASCII value of character at index 8: " + ascii);
    }
}
```

Output :

Original Text: Object Oriented Programming

The given text is empty -> false

Length of the given text: 29

To Uppercase: OBJECT ORIENTED PROGRAMMING

To Lowercase: object oriented programming

Substring: Oriented Programming

Subsequence: Oriented

Replace O with \$: \$bject \$riented Programming

Replace all text with 'Java Lab': Java Lab

Replace first word: Java Oriented Programming

Character at index 8: r

ASCII value of character at index 8: 114

Experiment - 9**Aim :**

Program to create a class for Employee and Search for an employee given eNo, using the concept of Array of Objects.

Program Code :

```
import java.util.Scanner;

class Employee {
    public String eNo;
    public String eName;
    public int eSalary;
    public Employee(String eNo,String eName,int eSalary) {
        this.eNo = eNo;
        this.eName = eName;
        this.eSalary = eSalary;
    }
}

public class mainEmployee {
    public static void main(String args[]) {
        Scanner scan = new Scanner(System.in);
        System.out.println("Enter count of employee list:");
        int count = Integer.parseInt(scan.nextLine());
        System.out.println();
        Employee[] employees = new Employee[count];
        for(int i=0;i<count;i++)
        {
            System.out.print("Enter number of employee:");
            String eNo = scan.nextLine();
            System.out.print("Enter name of employee:");
            String eName = scan.nextLine();
            System.out.print("Enter salary of employee:");
            int eSalary = Integer.parseInt(scan.nextLine());
            System.out.println();
            employees[i] = new Employee(eNo,eName,eSalary);
        }
    }
}
```

```
    }  
    System.out.println("Enter the number of employee to be filtered out:");  
    String search = scan.nextLine();  
    System.out.println();  
    for(int i=0;i<count;i++) {  
        if(employees[i].eNo.equals(search)) {  
            System.out.println("Employee number:" +employees[i].eNo);  
            System.out.println("Employee name:" +employees[i].eName);  
            System.out.println("Employee salary:" +employees[i].eSalary);  
        }  
    }  
}
```

Output :

Enter count of employee list:

2

Enter number of employee:001

Enter name of employee:aarathy

Enter salary of employee:50000

Enter number of employee:004

Enter name of employee:saniya

Enter salary of employee:65000

Enter the number of employee to be filtered out:

004

Employee number:004

Employee name:saniya

Employee salary:65000

Experiment - 10**Aim :**

Area of different shapes using overloaded functions.

Program Code :

```
import java.util.*;

public class areaComputer {

    int a;

    int area;

    float x;

    double y;

    void area(int l,int b) {

        int length = l;

        int breadth = b;

        a = length * breadth;

        System.out.println("Area: " +a);

    }

    void area(int s) {

        int side = s;

        area = side * side;

        System.out.println("Area: " +area);

    }

    void area(float b,float h) {

        float base = b;

        float height = h;

        x=(base*height)/2;

        System.out.println("Area: " +x);

    }

    void area(double r) {

        double radius = r;

        y = 3.14*radius*radius;

        System.out.println("Area: "+y);

    }

}
```

```

public static void main(String args[]) {
    areaComputer obj = new areaComputer();
    Scanner sc = new Scanner(System.in);
    System.out.println("***** Area of different shapes using overloaded functions
*****");

    System.out.println("\n ---- RECTANGLE ----");
    System.out.println("Enter the length and breadth:");
    int l = sc.nextInt();
    int b = sc.nextInt();
    obj.area(l,b);
    System.out.println("\n ---- SQUARE ----");
    System.out.println("Enter the side:");
    int s = sc.nextInt();
    obj.area(s);
    System.out.println("\n ---- TRIANGLE ----");
    System.out.println("Enter the base and height:");
    float base = sc.nextFloat();
    float height = sc.nextFloat();
    obj.area(base,height);
    System.out.println("\n ---- CIRCLE ----");
    System.out.println("Enter the radius:");
    double r = sc.nextDouble();
    obj.area(r);
    sc.close();
}
}

```

Output :

***** Area of different shapes using overloaded functions *****

---- RECTANGLE ----

Enter the length and breadth:

3

4

Area: 12

---- SQUARE ----

Enter the side:

4

Area: 16

---- TRIANGLE ----

Enter the base and height:

3

6

Area: 9.0

---- CIRCLE ----

Enter the radius:

3

Area: 28.26

Experiment - 11**Aim :**

Create a class 'Employee' and another class 'Teacher' that inherit the properties of class .Use array of objects to display details of N teachers.

Program Code :

```
import java.util.Scanner;

class Employee {
    int Empid;
    String Name;
    double Salary;
    String Address;
    Employee(int no,String na,double sal,String add) {
        this.Empid = no;
        this.Name = na;
        this.Salary = sal;
        this.Address = add;
    }
}

class Teacher extends Employee {
    String dept;
    String subject;
    Teacher(int no,String na,double sal,String add,String dep,String sub) {
        super(no,na,sal,add);
        this.dept = dep;
        this.subject = sub;
    }
    void display() {
        System.out.println("Employee id: "+Empid);
        System.out.println("Name: "+Name);
        System.out.println("Salary: "+Salary);
        System.out.println("Address: "+Address);
        System.out.println("Department: "+dept);
        System.out.println("Subject: "+subject);
    }
}
```

```

    }

    public static void main(String args[]) {
        System.out.println("Enter the no. of employees: ");
        Scanner sc1 = new Scanner(System.in);
        int num = sc1.nextInt();
        Teacher arr[] = new Teacher[num];
        for(int i=0;i<num;i++) {
            Scanner sc = new Scanner(System.in);
            System.out.println("Enter employee id:");
            int Empid = sc.nextInt();
            System.out.println("Enter employee name:");
            String Name = sc.next();
            System.out.println("Enter salary:");
            double Salary = sc.nextDouble();
            sc.nextLine();
            System.out.println("Enter Address:");
            String Address = sc.nextLine();
            System.out.println("Enter department:");
            String dept = sc.nextLine();
            System.out.println("Enter subject:");
            String subject = sc.next();
            arr[i] = new Teacher(Empid,Name,Salary,Address,dept,subject);
        }
        System.out.println("\n *** Informations of all the employees ***");
        for(int i=0;i<num;i++) {
            int j=i+1;
            System.out.println("\n" +j+ ".");
            arr[i].display();
        }
        sc1.close();
    }
}

```

Output :

Enter the no. of employees:

2

Enter employee id:

101

Enter employee name:

Saniya

Enter salary:

50000

Enter Address:

abc house

Enter department:

CA

Enter subject:

AOS

Enter employee id:

102

Enter employee name:

Dina

Enter salary:

60000

Enter Address:

xyz house

Enter department:

MECH

Enter subject:

Thermodynamics

*** Informations of all the employees ***

1).

Employee id: 101

Name: Saniya

Salary: 50000.0

Address: abc house

Department: CA

Subject: AOS

2).

Employee id: 102

Name: Dina

Salary: 60000.0

Address: xyz house

Department: MECH

Subject: Thermodynamics

Experiment - 12**Aim :**

Create a class 'Person' and another class 'Employee' that inherits the properties of class Person .Use array of objects to display details of N teachers.

Program Code :

```
import java.util.Scanner;

class person {
    String name;
    String gender;
    String address;
    int age;
    person(String name,String gender,String address,int age) {
        this.name = name;
        this.gender = gender;
        this.address = address;
        this.age = age;
    }
}

class employee extends person {
    int empid;
    String company_name;
    String qualification;
    long salary;
    employee(String name,String gender,String address,int age,int empid,String
company_name,String qualification,long salary) {
        super(name,gender,address,age);
        this.empid = empid;
        this.company_name = company_name;
        this.qualification = qualification;
        this.salary = salary;
    }
}

public class teacher2 extends employee {
    String subject;
```



```

String department;

String teacher_id;

teacher2(String name,String gender,String address,int age,int empid,String
company_name,String qualification,long salary,String subject,String department,String teacher_id)
{

    super(name,gender,address,age,empid,company_name,qualification,salary);

    this.subject = subject;

    this.department = department;

    this.teacher_id = teacher_id;

}

void display() {

    System.out.println("Name:" +name);

    System.out.println("Gender:" +gender);

    System.out.println("Address:" +address);

    System.out.println("Age:" +age);

    System.out.println("Employee_id:" +empid);

    System.out.println("Company_name:" +company_name);

    System.out.println("Qualification:" +qualification);

    System.out.println("Salary:" +salary);

    System.out.println("Subject:" +subject);

    System.out.println("Department:" +department);

    System.out.println("Teacher_id:" +teacher_id);

}

public static void main(String args[]) {

    System.out.println("Enter the no.of teachers:");

    Scanner sc1 = new Scanner(System.in);

    int num = sc1.nextInt();

    teacher2 arr[] = new teacher2[num];

    System.out.println("\n");

    System.out.println("Enter the details of teachers:");

    int x=0,j=0;

    Scanner sc = new Scanner(System.in);

    for(int i=0;i<num;i++) {

        x=i+1;

```

```

        System.out.println("\n"+x+").");
        System.out.println("Name:");
        String a = sc.next();
        System.out.println("Gender:");
        String b = sc.next();
        System.out.println("Address:");
        String c = sc.next();
        System.out.println("Age:");
        int d = sc.nextInt();
        System.out.println("Employee_id:");
        int e = sc.nextInt();
        System.out.println("Company_name:");
        String f = sc.next();
        System.out.println("Qualification:");
        String g = sc.next();
        System.out.println("Salary:");
        long h = sc.nextLong();
        System.out.println("Subject:");
        String k = sc.next();
        System.out.println("Department:");
        String l = sc.next();
        System.out.println("Teacher_id:");
        String n = sc.next();
        arr[i] = new teacher2(a,b,c,d,e,f,g,h,k,l,n);
    }
    sc.close();
    System.out.println("\n *** Information of all the Teachers ***");
    for(int i=0;i<num;i++) {
        j=i+1;
        System.out.println("\n" +j+").");
        arr[i].display();
    }
    sc1.close();}}

```

Output :

Enter the no.of teachers:

2

Enter the details of teachers:

1).

Name:

Maaya

Gender:

Female

Address:

abc

Age:

28

Employee_id:

109

Company_name:

TCS

Qualification:

Btech

Salary:

50000

Subject:

Java

Department:

CS

Teacher_id:

T8

2).

Name:

Jacob

Gender:

Male

Address:

xyz

Age:

31

Employee_id:

105

Company_name:

Wipro

Qualification:

mca

Salary:

65000

Subject:

Python

Department:

CA

Teacher_id:

T4

*** Information of all the Teachers ***

1).

Name:Maaya

Gender:Female

Address:abc

Age:28

Employee_id:109

Company_name:TCS

Qualification:Btech

Salary:50000

Subject:Java

Department:CS

Teacher_id:T8

2).

Name:Jacob

Gender:Male

Address:xyz

Age:31

Employee_id:105

Company_name:Wipro

Qualification:mca

Salary:65000

Subject:Python

Department:CA

Teacher_id:T4

Experiment - 13**Aim :**

Write a program has class Publisher, Book, Literature and Fiction. Read the information and print the details of books from either the category, using inheritance.

Program Code :

```
import java.util.Scanner;

class Publisher {
    String publisher;

    Publisher(String pub) {
        this.publisher = pub;
    }
}

class Book extends Publisher {
    String book;

    Book(String pub, String boo) {
        super(pub);
        this.book = boo;
    }
}

class Literature extends Book {
    Literature(String pub, String boo) {
        super(pub, boo);
    }

    void display() {
        System.out.println("Publisher : " + publisher);
        System.out.println("Book : " + book);
    }
}

class Fiction extends Book {
    Fiction(String pub, String boo) {
        super(pub, boo);
    }

    void display() {
```

```

        System.out.println("Publisher : " + publisher);
        System.out.println("Book : " + book);
    }
}

public class bookDetails {
    public static void main(String[] args) {
        Scanner sc1 = new Scanner(System.in);
        Scanner sc = new Scanner(System.in);
        System.out.println("\nEnter the No. of Literature Books");
        int num = sc1.nextInt();
        Literature arr[] = new Literature[num];
        System.out.println("\nEnter the Literature Book Details\n");
        for (int i = 0; i < num; i++) {
            System.out.println("\n" + (i + 1) + ".");
            System.out.println("Book : ");
            String boo = sc.next();
            System.out.println("Publisher : ");
            String pub = sc.next();
            arr[i] = new Literature(pub, boo);
        }
        System.out.println("\nEnter the No. of Fiction Books");
        int num1 = sc1.nextInt();
        Fiction arr1[] = new Fiction[num1];
        System.out.println("\nEnter the Fiction Book Details\n");
        for (int i = 0; i < num1; i++) {
            System.out.println("\n" + (i + 1) + ".");
            System.out.println("Book : ");
            String boo = sc.next();
            System.out.println("Publisher : ");
            String pub = sc.next();
            arr1[i] = new Fiction(pub, boo);
        }
        System.out.println("\n *** Informations of all the Literature Books *** ");
    }
}

```

```
for (int i = 0; i < num; i++) {  
    System.out.println("\n" + (i + 1) + ").");  
    arr[i].display();  
}  
System.out.println("\n *** Informations of all the Fiction Books *** ");  
for (int i = 0; i < num1; i++) {  
    System.out.println("\n" + (i + 1) + ").");  
    arr1[i].display();  
}  
sc.close();  
sc1.close();  
}  
}
```


Output :

Enter the No. of Literature Books

2

Enter the Literature Book Details

1).

Book :

Hamlet

Publisher :

Penguin

2).

Book :

Odyssey

Publisher :

Oxford

Enter the No. of Fiction Books

2

Enter the Fiction Book Details

1).

Book :

HarryPotter

Publisher :

Bloomsbury

2).

Book :

Narnia

Publisher :

HarperCollins

*** Informations of all the Literature Books ***

1).

Publisher : Penguin

Book : Hamlet

2).

Publisher : Oxford

Book : Odyssey

*** Informations of all the Fiction Books ***

1).

Publisher : Bloomsbury

Book : HarryPotter

2).

Publisher : HarperCollins

Book : Narnia

Experiment - 14**Aim :**

Create classes Student and Sports. Create another class Result inherited from Student and Sports. Display academic and sports score of a student.

Program Code :

```
import java.util.Scanner;

class sports{
    String sport;
    int Rating;
    sports(String spo, int ra){
        sport = spo;
        Rating = ra;
    }
}

class student extends sports{
    String Grade;
    double Overall_per;
    student(String spo, int ra,String gd, double per ){
        super(spo, ra);
        Grade = gd;
        Overall_per = per;
    }
}

public class result extends student {
    result(String spo, int ra,String gd, double per ){
        super(spo, ra, gd, per);
    }
    void display(){
        System.out.println("\nSports Details of Student");
        System.out.println("Sport :"+sport);
        System.out.println("Rating :"+Rating);
        System.out.println("\nAcademic Details of Student");
        System.out.println("Academic Grade :"+Grade);
    }
}
```

```
        System.out.println("Overall percentage :"+Overall_per);
    }
    public static void main(String[] args) {
        Scanner sc =new Scanner(System.in);
        System.out.println("\nEnter the Sports Details of Student");
        System.out.println("\n Sport: ");
        String a =sc.next();
        System.out.println("\n Sport Rating  out of 10: ");
        int b =sc.nextInt();
        System.out.println("\nEnter the Academic Details of Student");
        System.out.println("\n Academic Grade: ");
        String c =sc.next();
        System.out.println("\n Overall percentage: ");
        double d =sc.nextDouble();
        sc.close();
        result obj= new result(a,b,c,d);
        obj.display();
    }
}
```

Output :

Enter the Sports Details of Student

Sport:

Cricket

Sport Rating out of 10:

8

Enter the Academic Details of Student

Academic Grade:

A

Overall percentage:

85.5

Sports Details of Student

Sport :Cricket

Rating :8

Academic Details of Student

Academic Grade :A

Overall percentage :85.5

Experiment - 15**Aim :**

Create an interface having prototypes of functions area() and perimeter(). Create a menu driven program to find area and perimeter of objects.

Program Code :

```
import java.util.Scanner;

interface prop
{
    void getdata();
    void area();
    void perimeter();
}

class Circle implements prop {
    double pi = 3.14;
    double r;
    Scanner sc = new Scanner(System.in);
    public void getdata() {
        System.out.println("Enter the radius of the circle: ");
        r = sc.nextDouble();
    }
    public void area() {
        System.out.println("Area of the circle: "+(pi*r*r));
    }
    public void perimeter() {
        System.out.println("Perimeter of the circle: "+(2*pi*r));
    }
}

class Rectangle implements prop {
    double l,b;
    Scanner sc = new Scanner(System.in);
    public void getdata() {
        System.out.println("Enter the length of the rectangle: ");
        l = sc.nextDouble();
        System.out.println("Enter the breadth of the rectangle: ");
```

```

        b = sc.nextDouble();
    }
    public void area() {
        System.out.println("Area of the rectangle: "+(l*b));
    }
    public void perimeter() {
        System.out.println("Perimeter of the rectangle: "+(2*(l+b)));
    }
}
public class prototype {
    public static void main(String args[]) {
        int ch;
        Scanner sc = new Scanner(System.in);
        Circle ob = new Circle();
        Rectangle obj = new Rectangle();
        do {
            System.out.println("\n1.Circle\n2.Rectangle\n3.Exit");
            System.out.println("Enter your choice:");
            ch = sc.nextInt();
            switch(ch) {
                case 1 :ob.getdata();
                        ob.area();
                        ob.perimeter();
                        break;
                case 2 :obj.getdata();
                        obj.area();
                        obj.perimeter();
                        break;
                case 3 :System.out.println("Exited...");
                        System.exit(0);
            }
        }
        while(true);
    }
}

```

Output :

1.Circle

2.Rectangle

3.Exit

Enter your choice:

1

Enter the radius of the circle:

3

Area of the circle: 28.26

Perimeter of the circle: 18.84

1.Circle

2.Rectangle

3.Exit

Enter your choice:

2

Enter the length of the rectangle:

4

Enter the breadth of the rectangle:

5

Area of the rectangle: 20.0

Perimeter of the rectangle: 18.0

1.Circle

2.Rectangle

3.Exit

Enter your choice:

3

Exited...

Experiment - 16**Aim :**

Prepare bill with the given format using calculate method from interface.

Program Code :

```
import java.util.Scanner;

interface calc {

    void calculate();

}

class bill implements calc {

    String date,name,p_id;

    int quantity;

    double unit_price,total,namount=0;

    Scanner sc = new Scanner(System.in);

    public void getdata() {

        System.out.println("\nEnter product id:");

        p_id = sc.nextLine();

        System.out.println("Enter product name:");

        name = sc.nextLine();

        System.out.println("Enter the Quantity:");

        quantity = sc.nextInt();

        System.out.println("Enter the unit price:");

        unit_price = sc.nextDouble();

    }

    public void calculate() {

        total = quantity * unit_price;

    }

    public void display() {

        System.out.println(p_id+"\t\t"+name+"\t\t"+quantity+"\t\t"+unit_price+"\t"+total);

    }

}

public class billMain {

    public static void main(String[] args) {

        int n,i;
```

```

        double namount=0,t;
        int ran;
        String date;
        t = Math.random() *1000000;
        ran = (int) t;
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the date:");
        date = sc.nextLine();
        System.out.println("Enter how many products are there:");
        n = sc.nextInt();
        bill ob[] = new bill[n];
        for(i=0;i<n;i++) {
            ob[i] = new bill();
        }
        for(i=0;i<n;i++) {
            ob[i].getdata();
            ob[i].calculate();
        }
        System.out.println("\n");
        System.out.println("Order no. #"+ran);
        System.out.println("Date:"+date);
        System.out.println("Product Id \tName\t Quantity\t unit price\t Total ");
        System.out.println("-----");
        for(i=0;i<n;i++) {
            ob[i].display();
            namount += ob[i].total;
        }
        System.out.println("-----");
        System.out.println("\t\t\tNet.Amount\t"+ namount);
    }
}

```

Output :

Enter the date:

12-03-2026

Enter how many products are there:

2

Enter product id:

p1

Enter product name:

Soap

Enter the Quantity:

50

Enter the unit price:

25

Enter product id:

p2

Enter product name:

Brush

Enter the Quantity:

36

Enter the unit price:

15

Order no. #935840

Date: 12-03-2026

Product Id	Name	Quantity	Unit Price	Total
------------	------	----------	------------	-------

p1	Soap	50	25.0	1250.0
----	------	----	------	--------

p2	Brush	36	15.0	540.0
----	-------	----	------	-------

Net Amount: 1790.0

Experiment - 17**Aim :**

Create a Graphics package that has classes and interfaces for figures Rectangle, Triangle, Square and Circle. Test the package by finding the area of these figures basic arithmetic operations. Test the package by implementing all.

Program Code :

i)

```
import java.util.Scanner;

interface Figures {

    void rectangle();

    void triangle();

    void square();

    void circle();

}

public class Area implements Figures {

    Scanner s = new Scanner(System.in);

    public void rectangle() {

        System.out.print("Enter the length: ");

        int l = s.nextInt();

        System.out.print("Enter the breadth: ");

        int b = s.nextInt();

        System.out.println("Area of the rectangle: " + (l * b));

    }

    public void triangle() {

        System.out.print("Enter the base: ");

        int base = s.nextInt();

        System.out.print("Enter the height: ");

        int h = s.nextInt();

        System.out.println("Area of the triangle: " + (0.5 * base * h));

    }

    public void square() {

        System.out.print("Enter the side: ");

        int side = s.nextInt();

        System.out.println("Area of the square: " + (side * side));

    }

}
```

```

    }

    public void circle() {
        System.out.print("Enter the radius: ");
        float r = s.nextFloat();
        System.out.println("Area of the circle: " + (Math.PI * r * r));
    }

    public static void main(String[] args) {
        Area obj = new Area();
        obj.rectangle();
        obj.triangle();
        obj.square();
        obj.circle();
    }
}

ii)
import Graphics.Area;

public class test_package {

    public static void main(String[] args) {
        Area A = new Area();
        System.out.println("**COMPUTATION OF AREA OF DIFFERENT FIGURES**");
        System.out.println("\n__RECTANGLE__");
        A.rectangle();
        System.out.println("\n__TRIANGLE__");
        A.triangle();
        System.out.println("\n__SQUARE__");
        A.square();
        System.out.println("\n__CIRCLE__");
        A.circle();
    }
}

```

Output :

****COMPUTATION OF AREA OF DIFFERENT FIGURES****

__RECTANGLE__

Enter the length: 4

Enter the breadth: 5

Area of the rectangle: 20

__TRIANGLE__

Enter the base: 6

Enter the height: 3

Area of the triangle: 9.0

__SQUARE__

Enter the side: 4

Area of the square: 16

__CIRCLE__

Enter the radius: 2

Area of the circle: 12.56

Experiment - 18**Aim :**

Create an Arithmetic package that has classes and interfaces for the 4 basic arithmetic operations. Test the package by implementing all operations on two given numbers.

Program Code :**i)**

```
import java.util.Scanner;
import ArithmeticPackage.Operands;
public class MathOperations {
    public static void main(String[] args) {
        Scanner s = new Scanner(System.in);
        float a, b;
        System.out.println("____ARITHMETIC OPERATIONS____");
        System.out.print("Enter any two numbers to be operated with: ");
        a = s.nextFloat();
        b = s.nextFloat();
        Operands op = new Operands(a, b);
        op.add();
        op.sub();
        op.mul();
        op.div();
        s.close();
    }
}
```

ii)

```
package ArithmeticPackage;
interface Operations {
    void add();
    void sub();
    void div();
    void mul();
}
public class Operands implements Operations {
```

```
float x, y;

public Operands(float a, float b) {
    x = a;
    y = b;
}

public void add() {
    System.out.println("\nSUM = " + (x + y));
}

public void sub() {
    System.out.println("\nDIFFERENCE = " + (x - y));
}

public void div() {
    System.out.println("\nQUOTIENT = " + (x / y));
}

public void mul() {
    System.out.println("\nPRODUCT = " + (x * y));
}
}
```


Output :

____ARITHMETIC OPERATIONS____

Enter any two numbers to be operated with: 10 5

SUM = 15.0

DIFFERENCE = 5.0

QUOTIENT = 2.0

PRODUCT = 50.0

Experiment - 19**Aim :**

Write a user defined exception class to authenticate the user name and password.

Program Code :

i)

```
import java.util.Scanner;

public class authentication {

    public static void main(String args[]) {

        String username = "MBITS";
        String password = "MBITS";
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the username:");
        String u1 = sc.nextLine();
        System.out.println("Enter the password:");
        String u2 = sc.nextLine();

        try {

            if((u1.equals(username)) && (u2.equals(password))) {

                System.out.println("Access Granted");

            }

            else {

                throw new credentialexception("Invalid Credentials");

            }

        }

        catch(credentialexception e) {

            System.out.println(e.getMessage());

        }

    }

}
```

ii)

```
public class credentialexception extends Exception {

    public credentialexception(String s) {

        super(s);

    }

}
```

Output :

Enter the username:

MBITS

Enter the password:

MBITS

Access Granted

Experiment - 20**Aim :**

Find the average of N positive integers, raising a user defined exception for each negative input.

Program Code :

i)

```
import java.util.Scanner;

public class average {

    public static void main(String args[]) {

        double sum = 0;

        Scanner sc = new Scanner(System.in);

        System.out.println("Count of numbers:");

        int N = sc.nextInt();

        int[] numbers = new int[N];

        for(int i=0;i<N;i++) {

            System.out.println("Enter the number:");

            numbers[i] = sc.nextInt();

        }

        for(int i=0;i<N;i++) {

            try {

                if(numbers[i]>=0) {

                    sum+=numbers[i];

                }

                else {

                    throw new negative_exception("Negative number:" +numbers[i]);

                }

            }

            catch(negative_exception e) {

                System.out.println(e.getMessage());

            }

        }

        double avg = sum/N;

        System.out.println("Average of positive numbers = " +avg);

    }

}
```

```
    }  
}  
ii)  
public class negative_exception extends Exception {  
    public negative_exception(String s) {  
        super(s);  
    }  
}
```

Output :

Count of numbers:

3

Enter the number:

-12

Enter the number:

23

Enter the number:

34

Negative number:-12

Average of positive numbers = 19.0

Experiment - 21**Aim :**

Define 2 classes; one for generating multiplication table of 5 and other for displaying first N prime numbers. Implement using threads. (Thread class)

Program Code :

i)

```
public class Prime extends Thread {
    int num;

    public Prime(int n) {
        this.num = n;
    }

    public void run() {
        int x, y, flag;

        System.out.println("All the Prime numbers within 1 and " + num + " are:");

        for (x = 1; x <= num; x++) {
            if (x == 1 || x == 0)
                continue;

            flag = 1;

            for (y = 2; y <= x / 2; y++) {
                if (x % y == 0) {
                    flag = 0;
                    break;
                }
            }

            if (flag == 1)
                System.out.println("Prime number = " + x);
        }
    }
}
```

ii)

```
public class Mul extends Thread {
    public void run() {
        for (int i = 0; i <= 10; i++) {
```

```
        System.out.println("5 * " + i + " = " + (5 * i));  
    }  
}  
}
```

iii)

```
import java.util.Scanner;  
  
public class Driver {  
    public static void main(String[] args) {  
        System.out.println("Enter the number:");  
        Scanner sc = new Scanner(System.in);  
        int number = sc.nextInt();  
        Mul obj1 = new Mul();  
        obj1.start();  
        Prime obj2 = new Prime(number);  
        obj2.start();  
    }  
}
```


Output :

Enter the number:

3

All the Prime numbers within 1 and 3 are:

Prime number = 2

Prime number = 3

$$5 * 0 = 0$$

$$5 * 1 = 5$$

$$5 * 2 = 10$$

$$5 * 3 = 15$$

$$5 * 4 = 20$$

$$5 * 5 = 25$$

$$5 * 6 = 30$$

$$5 * 7 = 35$$

$$5 * 8 = 40$$

$$5 * 9 = 45$$

$$5 * 10 = 50$$

Experiment - 22**Aim :**

Define 2 classes; one for generating Fibonacci numbers and other for displaying even numbers in a given range. Implement using threads.(Runnable Interface)

Program Code :

i)

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter the value of n for Fibonacci series: ");

        int n = scanner.nextInt();

        System.out.print("Enter the range (start end): ");

        int start = scanner.nextInt();

        int end = scanner.nextInt();

        EvenNumberPrinter evenNumberPrinter = new EvenNumberPrinter(start, end);

        Thread evenThread = new Thread(evenNumberPrinter);

        evenThread.start();

        FibonacciGenerator fibonacciGenerator = new FibonacciGenerator(n);

        Thread fibonacciThread = new Thread(fibonacciGenerator);

        fibonacciThread.start();

    }

}
```

ii)

```
public class FibonacciGenerator implements Runnable {

    private int n;

    public FibonacciGenerator(int n) {

        this.n = n;

    }

    @Override

    public void run() {

        int a = 0, b = 1;

        System.out.println("fibonacci = " + a);

    }

}
```

```

        System.out.println("fibonacci = " + b);
    for (int i = 2; i < n; i++) {
        int c = a + b;
        System.out.println("fibonacci = " + c);
        a = b;
        b = c;
    }
}
}

iii)
public class EvenNumberPrinter implements Runnable {
    private int start;
    private int end;
    public EvenNumberPrinter(int start, int end) {
        this.start = start;
        this.end = end;
    }
    @Override
    public void run() {
        for (int i = start; i <= end; i++) {
            if (i % 2 == 0) {
                System.out.println("Even = " + i);
            }
        }
    }
}

```

Output :

Enter the value of n for Fibonacci series: 5

Enter the range (start end):

1

10

Even = 2

Even = 4

fibonacci = 0

fibonacci = 1

Even = 6

fibonacci = 1

Even = 8

fibonacci = 2

Even = 10

fibonacci = 3

Experiment - 23**Aim :**

Producer/Consumer using IPC.

Program Code :

```
class Buffer {  
    private int data;  
    private boolean available = false;  
    public synchronized void produce(int value) {  
        while (available) {  
            try {  
                wait();  
            } catch (InterruptedException e) {  
                System.out.println(e);  
            }  
        }  
        data = value;  
        System.out.println("Produced: " + data);  
        available = true;  
        notify();  
    }  
    public synchronized void consume() {  
        while (!available) {  
            try {  
                wait();  
            } catch (InterruptedException e) {  
                System.out.println(e);  
            }  
        }  
        System.out.println("Consumed: " + data);  
        available = false;  
        notify();  
    }  
}
```

```
class Producer extends Thread {  
    Buffer buffer;  
    Producer(Buffer buffer) {  
        this.buffer = buffer;  
    }  
    public void run() {  
        for (int i = 1; i <= 5; i++) {  
            buffer.produce(i);  
        }  
    }  
}  
  
class Consumer extends Thread {  
    Buffer buffer;  
    Consumer(Buffer buffer) {  
        this.buffer = buffer;  
    }  
    public void run() {  
        for (int i = 1; i <= 5; i++) {  
            buffer.consume();  
        }  
    }  
}  
  
public class ProducerConsumerIPC {  
    public static void main(String[] args) {  
        Buffer buffer = new Buffer();  
        Producer p = new Producer(buffer);  
        Consumer c = new Consumer(buffer);  
        p.start();  
        c.start();  
    }  
}
```

Output :

Produced: 1

Consumed: 1

Produced: 2

Consumed: 2

Produced: 3

Consumed: 3

Produced: 4

Consumed: 4

Produced: 5

Consumed: 5

Experiment - 24**Aim :**

Program to create a generic stack and do the Push and Pop operations.

Program Code :

```

class Stack<T> {
    private int size;
    private int top;
    private T[] arr;
    @SuppressWarnings("unchecked")
    Stack(int size) {
        this.size = size;
        this.top = -1;
        arr = (T[]) new Object[size];
    }
    public void push(T value) {
        if (top == size - 1) {
            System.out.println("Stack Overflow");
        } else {
            arr[++top] = value;
            System.out.println("Pushed: " + value);
        }
    }
    public void pop() {
        if (top == -1) {
            System.out.println("Stack Underflow");
        } else {
            System.out.println("Popped: " + arr[top--]);
        }
    }
    public void display() {
        if (top == -1) {
            System.out.println("Stack is empty");
        } else {

```



```
        System.out.print("Stack elements: ");
        for (int i = top; i >= 0; i--) {
            System.out.print(arr[i] + " ");
        }
        System.out.println();
    }
}

public class GenericStackDemo {
    public static void main(String[] args) {
        Stack<Integer> s = new Stack<>(5);
        s.push(10);
        s.push(20);
        s.push(30);
        s.display();
        s.pop();
        s.pop();
        s.display();
    }
}
```

Output :

Pushed: 10

Pushed: 20

Pushed: 30

Stack elements: 30 20 10

Popped: 30

Popped: 20

Stack elements: 10

Experiment - 25**Aim :**

Using generic method perform Bubble sort.

Program Code :

```

class GenericBubbleSort {
    public static <T extends Comparable<T>> void bubbleSort(T[] arr) {
        int n = arr.length;
        for (int i = 0; i < n - 1; i++) {
            for (int j = 0; j < n - i - 1; j++) {
                if (arr[j].compareTo(arr[j + 1]) > 0) {
                    T temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                }
            }
        }
    }

    public static <T> void display(T[] arr) {
        for (T item : arr) {
            System.out.print(item + " ");
        }
        System.out.println();
    }

    public static void main(String[] args) {
        Integer[] intArr = {5, 2, 9, 1, 6};
        System.out.println("Before Sorting:");
        display(intArr);
        bubbleSort(intArr);
        System.out.println("After Sorting:");
        display(intArr);
    }
}

```

Output :

Before Sorting:

5 2 9 1 6

After Sorting:

1 2 5 6 9

Experiment - 26**Aim :**

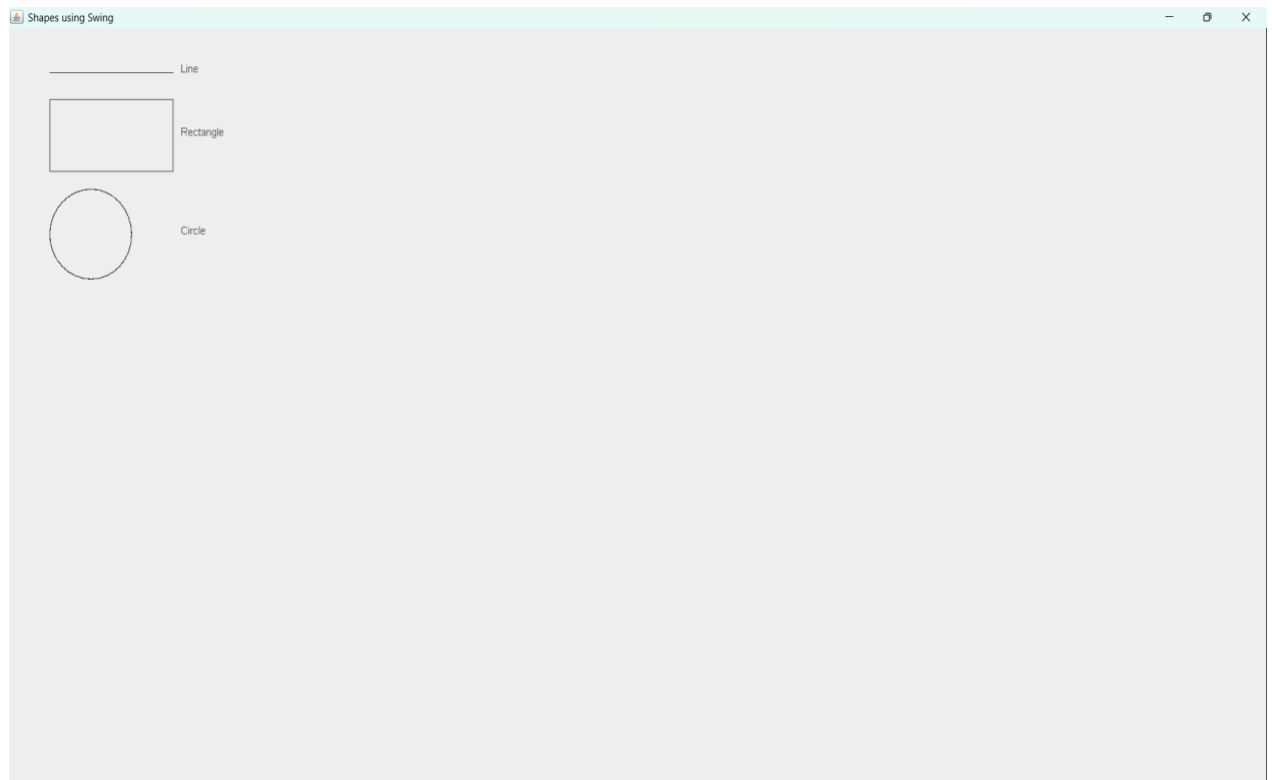
Program to draw Circle, Rectangle, Line in Applet.

Program Code :

```
import javax.swing.*;
import java.awt.*;

public class ShapeSwing extends JPanel {
    protected void paintComponent(Graphics g) {
        super.paintComponent(g);
        g.drawLine(50, 50, 200, 50);
        g.drawRect(50, 80, 150, 80);
        g.drawOval(50, 180, 100, 100);
        g.drawString("Line", 210, 50);
        g.drawString("Rectangle", 210, 120);
        g.drawString("Circle", 210, 230);
    }

    public static void main(String[] args) {
        JFrame frame = new JFrame("Shapes using Swing");
        ShapeSwing panel = new ShapeSwing();
        frame.add(panel);
        frame.setSize(400, 400);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```

Output :

Experiment - 27**Aim :**

Program to find maximum of three numbers using AWT.

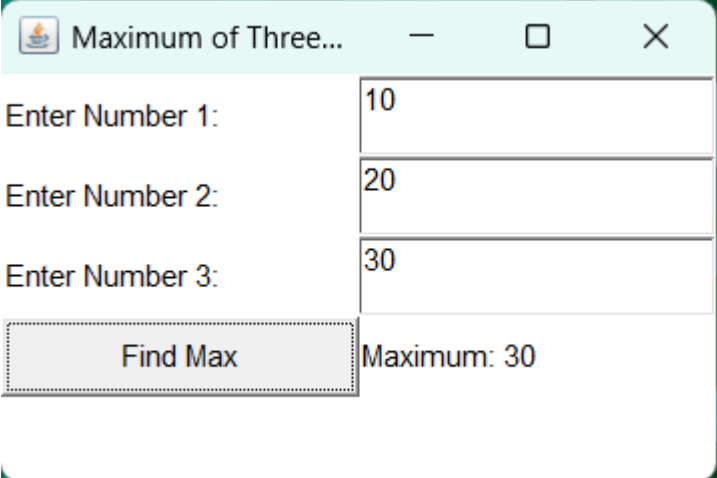
Program Code :

```
import java.awt.*;
import java.awt.event.*;

public class MaxAWT extends Frame implements ActionListener {
    Label l1, l2, l3, result;
    TextField t1, t2, t3;
    Button b;

    MaxAWT() {
        l1 = new Label("Enter Number 1:");
        l2 = new Label("Enter Number 2:");
        l3 = new Label("Enter Number 3:");
        result = new Label("Result:");
        t1 = new TextField();
        t2 = new TextField();
        t3 = new TextField();
        b = new Button("Find Max");
        b.addActionListener(this);
        setLayout(new GridLayout(5, 2));
        add(l1); add(t1);
        add(l2); add(t2);
        add(l3); add(t3);
        add(b); add(result);
        setTitle("Maximum of Three Numbers");
        setSize(300, 200);
        setVisible(true);
        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent we) {
                dispose();
            }
        });
    }
}
```

```
}  
  
public void actionPerformed(ActionEvent e) {  
    int n1 = Integer.parseInt(t1.getText());  
    int n2 = Integer.parseInt(t2.getText());  
    int n3 = Integer.parseInt(t3.getText());  
    int max = n1;  
    if (n2 > max) max = n2;  
    if (n3 > max) max = n3;  
    result.setText("Maximum: " + max);  
}  
  
public static void main(String[] args) {  
    new MaxAWT();  
}  
}
```


Output :

The screenshot shows a Java Swing window titled "Maximum of Three...". Inside the window, there are three input fields labeled "Enter Number 1:", "Enter Number 2:", and "Enter Number 3:". The values entered are 10, 20, and 30 respectively. Below these fields is a button labeled "Find Max". To the right of the button, the text "Maximum: 30" is displayed, indicating the result of the calculation.

Enter Number 1:	10
Enter Number 2:	20
Enter Number 3:	30
Find Max	Maximum: 30

Experiment - 28**Aim :**

Find the percentage of marks obtained by a student in 5 subjects. Display a happy face if he secures above 50% or a sad face if otherwise.

Program Code :

```
import java.awt.*;
import java.awt.event.*;

class FacePanel extends Panel {
    int percentage = -1;

    public void setPercentage(int p) {
        percentage = p;
        repaint();
    }

    public void paint(Graphics g) {
        if (percentage == -1) return;
        int x = 40, y = 10, size = 60;
        g.drawOval(x, y, size, size);
        g.fillOval(x + 15, y + 18, 6, 6);
        g.fillOval(x + 35, y + 18, 6, 6);
        if (percentage > 50) {
            g.drawArc(x + 15, y + 30, 30, 15, 180, 180);
        } else {
            g.drawArc(x + 15, y + 35, 30, 15, 0, 180);
        }
    }
}

public class PercentageFace extends Frame implements ActionListener {
    TextField t1, t2, t3, t4, t5;
    Label result;
    FacePanel facePanel;

    PercentageFace() {
        setLayout(new FlowLayout());
        add(new Label("Sub1:")); t1 = new TextField(5); add(t1);
        add(new Label("Sub2:")); t2 = new TextField(5); add(t2);
```

```

add(new Label("Sub3:")); t3 = new TextField(5); add(t3);
add(new Label("Sub4:")); t4 = new TextField(5); add(t4);
add(new Label("Sub5:")); t5 = new TextField(5); add(t5);
Button b = new Button("Calculate");
add(b);
b.addActionListener(this);
result = new Label("Result: ");
result.setPreferredSize(new Dimension(250, 20));
add(result);
facePanel = new FacePanel();
facePanel.setPreferredSize(new Dimension(120, 90));
add(facePanel);
setSize(300, 300); // compact window
setTitle("Percentage");
setVisible(true);
addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent we) {
        dispose();
    }
});
}
public void actionPerformed(ActionEvent e) {
    int m1 = Integer.parseInt(t1.getText());
    int m2 = Integer.parseInt(t2.getText());
    int m3 = Integer.parseInt(t3.getText());
    int m4 = Integer.parseInt(t4.getText());
    int m5 = Integer.parseInt(t5.getText());
    int percentage = (m1 + m2 + m3 + m4 + m5) / 5;
    result.setText("Percentage: " + percentage + "%");
    facePanel.setPercentage(percentage);
}
public static void main(String[] args) {
    new PercentageFace();
}
}

```

Output :

The screenshot shows a web application window titled "Percentage". Inside the window, there are five input fields labeled "Sub1:", "Sub2:", "Sub3:", "Sub4:", and "Sub5:". The values entered in these fields are 91, 89, 76, 82, and 99, respectively. To the right of these fields is a "Calculate" button. Below the input fields, the text "Percentage: 87%" is displayed. To the right of the "Calculate" button is a smiley face icon.

Sub	Value
Sub1	91
Sub2	89
Sub3	76
Sub4	82
Sub5	99

Calculate Percentage: 87%

Experiment - 29**Aim :**

Implement a simple calculator using AWT components.

Program Code :

```
import java.awt.*;
import java.awt.event.*;

public class CalculatorAWT extends Frame implements ActionListener {
    TextField t1, t2, result;
    Button add, sub, mul, div;

    CalculatorAWT() {
        setLayout(new GridLayout(5, 2, 5, 5)); // compact grid
        add(new Label("Num1:"));
        t1 = new TextField();
        add(t1);
        add(new Label("Num2:"));
        t2 = new TextField();
        add(t2);
        add = new Button("+");
        sub = new Button("-");
        mul = new Button("*");
        div = new Button("/");
        add(add); add(sub);
        add(mul); add(div);
        add.addActionListener(this);
        sub.addActionListener(this);
        mul.addActionListener(this);
        div.addActionListener(this);
        add(new Label("Result:"));
        result = new TextField();
        result.setEditable(false);
        add(result);

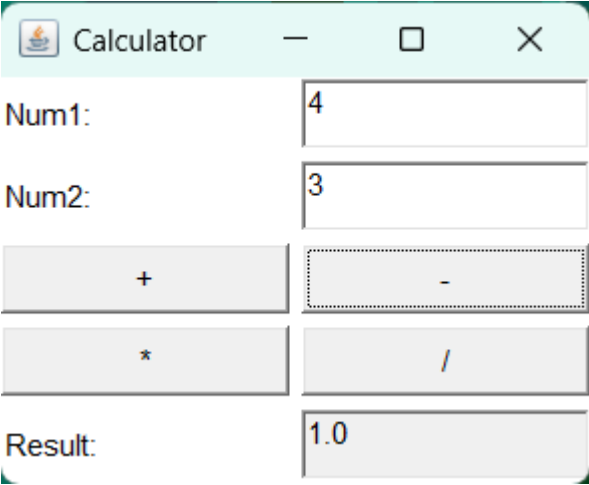
        setTitle("Calculator");
        setSize(250, 200);
    }
}
```

```
setVisible(true);
addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent we) {
        dispose();
    }
});
}

public void actionPerformed(ActionEvent e) {
    double n1 = Double.parseDouble(t1.getText());
    double n2 = Double.parseDouble(t2.getText());
    double res = 0;
    if (e.getSource() == add)
        res = n1 + n2;
    else if (e.getSource() == sub)
        res = n1 - n2;
    else if (e.getSource() == mul)
        res = n1 * n2;
    else if (e.getSource() == div)
        res = n1 / n2;
    result.setText(String.valueOf(res));
}

public static void main(String[] args) {
    new CalculatorAWT();
}
}
```

Output :



Calculator	
Num1:	4
Num2:	3
+	-
*	/
Result:	1.0

Experiment - 30**Aim :**

Program to list the sub directories and files in a given directory and also search for a file name.

Program Code :

```
import java.io.File;
import java.util.Scanner;

public class DirectorySearch {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter directory path: ");

        String path = sc.nextLine();

        File dir = new File(path);

        if (!dir.exists() || !dir.isDirectory()) {

            System.out.println("Invalid Directory");

            return;

        }

        System.out.println("\nFiles and Subdirectories:");

        File[] list = dir.listFiles();

        for (File f : list) {

            if (f.isDirectory()) {

                System.out.println("[DIR] " + f.getName());

            } else {

                System.out.println("[FILE] " + f.getName());

            }

        }

        System.out.print("\nEnter file name to search: ");

        String search = sc.nextLine();

        boolean found = false;

        for (File f : list) {

            if (f.getName().equals(search)) {

                found = true;

                break;

            }

        }

    }

}
```



```
    }  
    if (found) {  
        System.out.println("File Found");  
    } else {  
        System.out.println("File Not Found");  
    }  
    sc.close();  
}  
}
```

Output :

Enter directory path: D:\Test

Files and Subdirectories:

[DIR] Docs

[FILE] file1.txt

[FILE] program.java

Enter file name to search: file1.txt

File Found

Experiment - 31**Aim :**

Write a program to write to a file, then read from the file and display the contents on the console.

Program Code :

```
import java.io.FileWriter;
import java.io.FileReader;
import java.io.IOException;
import java.util.Scanner;

public class FileWriteRead {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        try {
            FileWriter fw = new FileWriter("sample.txt");
            System.out.print("Enter text to write into file: ");
            String text = sc.nextLine();
            fw.write(text);
            fw.close();
            System.out.println("\nData written to file successfully.");
            FileReader fr = new FileReader("sample.txt");
            int ch;
            System.out.println("\nReading from file:");
            while ((ch = fr.read()) != -1) {
                System.out.print((char) ch);
            }
            fr.close();
        } catch (IOException e) {
            System.out.println("Error occurred");
        }
        sc.close();
    }
}
```

Output :

Enter text to write into file: Hello Java File Handling

Data written to file successfully.

Reading from file:

Hello Java File Handling

Experiment - 32**Aim :**

Write a program to copy one file to another.

Program Code :

```
import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.IOException;
import java.util.Scanner;

public class FileCopy {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        try {
            System.out.print("Enter source file name: ");
            String source = sc.nextLine();
            System.out.print("Enter destination file name: ");
            String destination = sc.nextLine();
            FileInputStream fin = new FileInputStream(source);
            FileOutputStream fout = new FileOutputStream(destination);
            int ch;
            while ((ch = fin.read()) != -1) {
                fout.write(ch);
            }
            fin.close();
            fout.close();
            System.out.println("File copied successfully.");
        } catch (IOException e) {
            System.out.println("Error occurred");
        }
        sc.close();
    }
}
```

Output :

Enter source file name: source.txt

Enter destination file name: dest.txt

File copied successfully.

Experiment - 33**Aim :**

Write a program that reads from a file having integers. Copy even numbers and odd numbers to separate files.

Program Code :

```
import java.io.*;
import java.util.Scanner;
public class EvenOddFile {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        try {
            System.out.print("Enter input file name: ");
            String input = sc.nextLine();
            FileReader fr = new FileReader(input);
            BufferedReader br = new BufferedReader(fr);
            FileWriter evenWriter = new FileWriter("even.txt");
            FileWriter oddWriter = new FileWriter("odd.txt");
            String line;
            while ((line = br.readLine()) != null) {
                String[] nums = line.split(" ");
                for (String num : nums) {
                    int n = Integer.parseInt(num);
                    if (n % 2 == 0) {
                        evenWriter.write(n + " ");
                    } else {
                        oddWriter.write(n + " ");
                    }
                }
            }
            br.close();
            evenWriter.close();
            oddWriter.close();
            System.out.println("Even and Odd numbers separated successfully.");
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

```
    } catch (Exception e) {  
        System.out.println("Error occurred");  
    }  
    sc.close();  
}  
}
```


Output :

Enter input file name: input.txt

Even and Odd numbers separated successfully.

Contents of even.txt:

2 4 6 8 10

Contents of odd.txt:

1 3 5 7 9

Experiment - 34**Aim :**

Client server communication using Socket – TCP/IP.

Program Code :

i)

```
import java.io.*;
import java.net.*;

public class Server {
    public static void main(String[] args) {
        try {
            ServerSocket ss = new ServerSocket(5000);
            System.out.println("Server is waiting for client...");
            Socket s = ss.accept();
            System.out.println("Client connected");
            DataInputStream dis = new DataInputStream(s.getInputStream());
            DataOutputStream dos = new DataOutputStream(s.getOutputStream());
            String msg = dis.readUTF();
            System.out.println("Message from client: " + msg);
            dos.writeUTF("Message received by server");
            dis.close();
            dos.close();
            s.close();
            ss.close();
        } catch (Exception e) {
            System.out.println("Error occurred");
        }
    }
}
```

ii)

```
import java.io.*;
import java.net.*;
import java.util.Scanner;

public class Client {
```

```
public static void main(String[] args) {  
    Scanner sc = new Scanner(System.in);  
    try {  
        Socket s = new Socket("localhost", 5000);  
        DataOutputStream dos = new DataOutputStream(s.getOutputStream());  
        DataInputStream dis = new DataInputStream(s.getInputStream());  
        System.out.print("Enter message: ");  
        String msg = sc.nextLine();  
        dos.writeUTF(msg);  
        String reply = dis.readUTF();  
        System.out.println("Server reply: " + reply);  
        dos.close();  
        dis.close();  
        s.close();  
    } catch (Exception e) {  
        System.out.println("Error occurred");  
    }  
    sc.close();  
}
```

Output :

---- Server Side ----

Server is waiting for client...

Client connected

Message from client: Hello Server

---- Client Side ----

Enter message: Hello Server

Server reply: Message received by server

Experiment - 35**Aim :**

Client Server communication using Datagram Socket - UDP.

Program Code :

i)

```
import java.net.*;

public class UDPServer {
    public static void main(String[] args) {
        try {
            DatagramSocket ds = new DatagramSocket(5000);
            System.out.println("Server is waiting...");
            byte[] receive = new byte[1024];
            DatagramPacket dp = new DatagramPacket(receive, receive.length);
            ds.receive(dp);
            String msg = new String(dp.getData(), 0, dp.getLength());
            System.out.println("Message from client: " + msg);
            String reply = "Message received by server";
            byte[] send = reply.getBytes();
            DatagramPacket dpSend = new DatagramPacket(
                send, send.length, dp.getAddress(), dp.getPort()
            );
            ds.send(dpSend);
            ds.close();
        } catch (Exception e) {
            System.out.println("Error occurred");
        }
    }
}
```

ii)

```
import java.net.*;
import java.util.Scanner;

public class UDPClient {
    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);  
try {  
    DatagramSocket ds = new DatagramSocket();  
    System.out.print("Enter message: ");  
    String msg = sc.nextLine();  
    byte[] send = msg.getBytes();  
    InetAddress ip = InetAddress.getByName("localhost");  
    DatagramPacket dp = new DatagramPacket(  
        send, send.length, ip, 5000  
    );  
    ds.send(dp);  
    byte[] receive = new byte[1024];  
    DatagramPacket dpReceive = new DatagramPacket(receive, receive.length);  
    ds.receive(dpReceive);  
    String reply = new String(  
        dpReceive.getData(), 0, dpReceive.getLength()  
    );  
    System.out.println("Server reply: " + reply);  
    ds.close();  
} catch (Exception e) {  
    System.out.println("Error occurred");  
}  
sc.close();  
}
```

Output :

---- Server Side ----

Server is waiting...

Message from client: Hello UDP

---- Client Side ----

Enter message: Hello UDP

Server reply: Message received by server